higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## QUANTITY SURVEYING N6

## 14 AUGUST 2019

**This marking guideline consists of 8 pages.**

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## QUESTION 1

1.1 1.1.1 Number
1.1.2 Tons
1.1.3 Square metres (m²)
1.1.4 Cubic metres (m³)
1.1.5 Square metres (m²)
1.1.6 Metres (m)

(6 × 1) (6)

1.2 1.2.1 Drawn up by the measurer (taker-off)✓ of the work section that needs to be measured✓

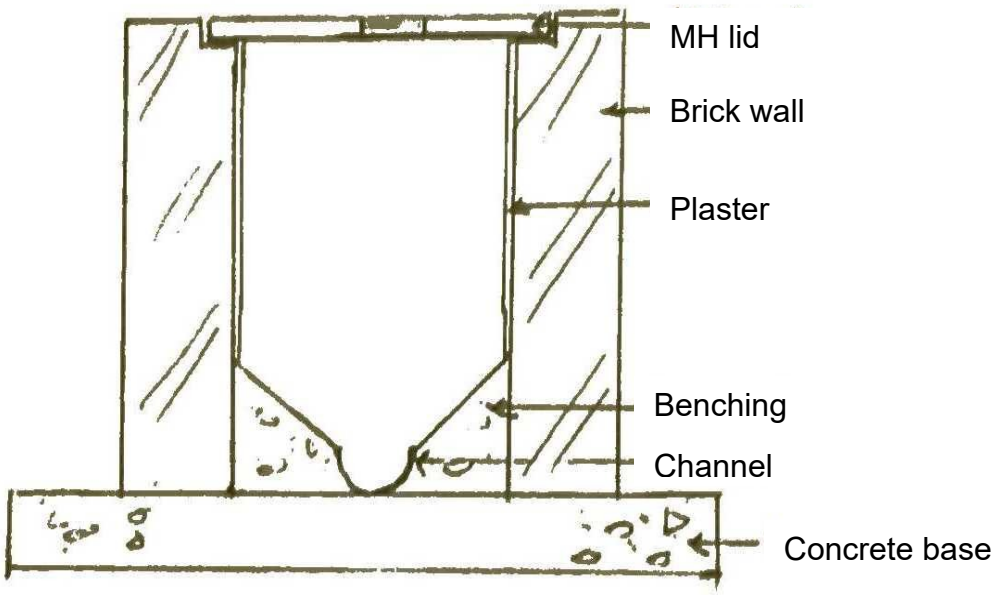
1.2.2 Rounding off✓ of decimal quantities to the nearest full quantity.✓

(2 × 2) (4)

1.3 1.3.1 Amount of money allowed in the bill of quantities.✓ for specialist work to be carried out by a specialist subcontractor✓ on a supply and fixing of material contract.✓ The final cost for the work will be established by the quantity surveyor✓ who will adjust the amount originally allowed in the bill of quantities. (5)

1.3.2 Amount of money✓allowed in the bill of quantities. For the supply only✓ of specialist materials or units✓ by a specialist supplier.✓ The final cost is adjusted by the quantity surveyor in the final account.✓ (5)

1.4




(5)
**[25]**

**QUESTION 2**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 5,38<br>0,60<br>0,52 | ✓ | Excav. in firm grd. for s.t. n.e. 2m dp. from g.l.✓<br>0,600 - 0,220 = 0,380 ÷ 2<br>= 0,190 m✓<br>5,000 + 2/0,190<br>=5,380 m✓<br>0,300 + 0,220= 0,520 m | | 3,50<br>5,00<br>0,08 | ✓ | 15Mpa. mass conc. in s.b incl. s.o. & c. hor. top surface.✓ |
| | | | | | 5,00<br>2,40 | ✓ | C. pls. bagging to int. vert. bk. wl.✓ |
| 2 | 5,38<br>0,52<br>0,60<br>0,52 | ✓<br>✓ | R.o.c. to sides of s.t. n.e. 1,5m dp.✓ | 2 | 5,00<br>2,63<br>0,22<br>2,63 | ✓<br>✓ | 19mm Thk. pls. to ext. vert bk wl.✓<br>2,400 + 0,075 + 2/75<br>= 2,625 m✓ |
| | 5,00<br>2,78 | ✓ | 220 mm Thk. bk. wl. in pls. bks. in 1:4 c.m. mix.✓<br>2,400 + 0,075 + 0300<br>=2,775m ✓ | | 5,00 | ✓ | 114 × 38 mm S.S.A.P. tbr. w.p.✓ |
| 2 | 5,38<br>0,19<br>0,30<br>0,22<br>0,19<br>0,30 | ✓<br>✓ | B/fill to sides of bk. wl. ✓ | | 5,00<br>0,30 | ✓ | Treat tbr. w.p. before fixing.✓<br>2/0,038 + 2/0,114<br>= 0,296 m✓ |
| | 3,50<br>5,00<br>0,15 | ✓ | Filling under conc. s.b.✓ | | | | |
| | | | (1) | | | | (2) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 5 | ✓ | Single mono plate-nailed tbr. roof truss, 3,5 m span, 0,5 m rise, two diag. struts and 300 mm eave projection.✓ | | 5,00<br>3,61 | ✓ | I.b.r. steel roof shts. ✓ |
| | 5 | ✓ | Wrought ends of tbr. rafters. ✓ | | 5,00 | ✓ | Waterproof flashing @ rf. & wl. intersection. ✓<br>&<br>250 × 38 mm W.S.A.P. tbr. facia brd. ✓<br>&<br>100 × 100 mm Pvc✓ Plastic incl. brackets screwed to tbr. fascia. |
| | 5 | ✓ | Extra for vert. plumb cut.✓ | | | | |
| | 5,00 | ✓ | 150 × 75 mm Tbr. w.p.✓ Incl. fixing steel brackets. | | 1 | ✓ | Extra for pvc outlet.✓ |
| 5 | 5,00 | ✓ | 76 × 50 mm Tbr. purlin on tbr. rafters.✓ | | 3,15 | ✓ | 75 Dia. pvc downpipe.✓<br>2,400<br>0,600 swanneck<br>O,150 shoe<br>3,150 m✓ |

(50 ÷ 2) **[25]**

## QUESTION 3

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | Door | | | | 2 Cts. clr. varnish to tbr. frame.✓ |
| | | | | | 2,88<br>0,27 | ✓ | Girth 2 x 76 = 0,152<br>+ 0,114<br>0,266 m✓ |
| | 1 | ✓ | 3-Pan. meranti dr. size 864 × 2 019 mm✓ | | | | |
| | | | 2,080 –(76-15) = 2,019 m✓<br>0,925-0,061 = 0,864 m ✓ | | | | Adjustments |
| | 0,91<br>2,06 | ✓ | 2 Cts. clr. varnish to tbr. dr.✓<br>2,019 × 0,864<br>+ 0,044 0,044<br>2,063 × 0,908 ✓ | | | | Ddt |
| | | | | | 0,93<br>2,18 | ✓ | 270 mm Thk. cav. bk. wl. a.b.✓ |
| | 3 | ✓ | 100mm H.d brass butt hinge. ✓ | | | | 2,080 + 0,100 = 2,180 m✓ |
| | 1 | ✓ | 3-Lever mortise lock. ✓ | | | | & |
| | | | | | | | Ddt. |
| | | | Frame | | | | E.o ord. bkwrk. for fcgs. a.b.✓ |
| | 5,09 | ✓ | 76 × 114 mm meranti tbr. in dr.frame once reb. incl. bldg. in.✓<br>2 × 2,080 = 4,160<br>+ 0,925<br>5,085 m ✓<br>& | | | | Ddt |
| | | | | | 0,93<br>2,08 | ✓ | Int. pls. to vert. bk. wl. a.b. ✓<br>& |
| | | | Prime backs of frame. ✓ | | | | Ddt<br>2 Cts. pva pt. a.b.✓ |
| | | | (1) | | | | (2) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | Ddt. | | | | |
| | 0,93 | ✓ | 76 mm Meranti sktg.✓ | | 0,93<br>0,08 | ✓ | Ditto, but to opening head.✓ |
| | | | & | | | | |
| | | | Ddt. | 2 | | | |
| | | | 2 Cts. clr. varnish to tbr. sktg. ✓ | | 0,08<br>2,08 | ✓ | Pls. to vert. bk. wl. in n.w. n.e. 300 mm wide.✓ |
| | | | Add. | | | | & |
| | 0,93<br>0,27<br>0,10 | ✓ | 15 Mpa mass conc. in s.b.✓<br>a.b. in thres. | | | | 2 Cts. pva. pt. in n.w. n.e. 300 mm wide.✓ |
| | 0,93<br>0,27 | ✓ | Pls. screen in thres.a.b. ✓ | | 0,93<br>0,08 | ✓ | Pls. to hor. bk. soffit a.b. ✓ |
| | | | | | | | & |
| | | | | | | | 2 Cts. pva pt. a.b.✓ |
| 4 | 0,93 | ✓ | Brk. reinf. in bldg. solid cav. in bkwrk. over dr. opening. ✓ | 2 | 0,08 | ✓ | 76 mm Meranti sktg. in reveals.✓ |
| | | | | | | | & |
| 2 | 2,08 | ✓ | Bldg. solid cav. to sides of tbr. frame.✓ | | | | 2 Cts. clr. varnish to tbr. sktg.✓ |
| 2 | 2,08<br>0,08 | ✓ | E.o. ord. bkwrk. for fcgs. in reveals.✓ | | | | |
| | | | (3) | | | | (4) |

( 50 ÷ 2 ) **[25]**

**QUESTION 4.1** OMISSIONS

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 4 / 6 / 20,00 | | ✓ | 12 mm Dia. h.s. reinf.✓ | | | | |
| 4 / 19,80 | | ✓ | 25 Mpa reinf. conc. in 450 mm dia. pile.✓ | | | | |
| 5 / 24,30 | | ✓ | 25 Mpa reinf. conc. 710 mm dia. pile.✓ | | | | |
| | 4 | ✓ | Add. conc. a.b.in enlarged bulb 450 – 755 mm.✓ | | | | |
| | 5 | ✓ | Add. conc. in enlarged bulb 610 – 970 mm✓ | | | | |

(10)

**QUESTION 4.2** ADDITIONS

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ✓ 4 / 6 ✓ | 20,00 | ✓ | ✓ 16 mm Dia. hs reinf.✓ | | | | |
| ✓ 4 | 19,80 | ✓ | ✓ 30 Mpa. reinf. conc. in 450 mm dia. pile.✓ | | | | |
| ✓ 5 | 24,30 | ✓ | Ditto, ditto but in 710 mm dia. pile.✓ | | | | |
| | 4 | ✓ | Add. Conc. a.b. in enlarged bulb 450 – 755 mm dia.✓ | | | | |
| | 5 | ✓ | Add. Conc. a.b. in enlarged bulb 610 – 970 mm dia.✓ | | | | |

(15)
**[25]**

**TOTAL: 100**

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